



**BALL STATE**  
UNIVERSITY

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# **Module 4 Lecture - Discrete Random Variables and Distributions**

Introduction to Statistical Methods

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# 1 Overview and Introduction

## 1.1 Textbook Learning Objectives

- Recognize and understand discrete probability distribution functions, in general.
- Calculate and interpret expected values.
- Recognize the binomial probability distribution and apply it appropriately.
- Classify discrete word problems by their distributions.

## 1.2 Instructor Learning Objectives

- Refresh on understanding of behavior and characteristics of discrete variables
- Understand the how to write probability distribution function tables and use them in calculation
- Be able to continue to write coherent and readable notation for probability problems

## 1.3 Introduction

- We will be expanding on what we spoke about with \_\_\_\_\_ in the last lecture, but now explicitly applying those ideas to variables, starting with \_\_\_\_\_ numeric variables

 Discuss: Try describing discrete variables in your own words, based on what you know from Module 1

- We previously described **discrete variable** data, or that which is numeric and “count-able”, that has no intervals between integers.
  - Those will be the focus of this lecture, before we focus on \_\_\_\_\_ variables in the next lecture

**! Important**

It's vital that you can distinguish between a continuous and discrete numeric variable, as it will have an impact on what tests you can do with it!

- Discrete vs continuous
  - Examples of discrete variables:
    - \* Number of days a student attends class
    - \* Number of days a patient reports being “good” in the last week
    - \* Number of times a student acts out in a school days
  - Examples of continuous variables:
    - \* GPA
    - \* Age
    - \* Height
- In this lecture, we will introduce the concept of **random variables**, or those that can vary in subsequent                      (used in the sense as how it was introduced during the probability lecture)
- The                      of random variables is as follows:
  - Uppercase alphabetical character, e.g.  $X$ ,  $Y$ ,  $Z$ , etc. → written/verbose of variable
  - Lowercase alphabetical equivalent, e.g.  $x$ ,  $y$ ,  $z$ , etc. → the possible                       $X$  can take on
- Example from book:
  - $X$  = the number of heads you get when you toss three                      coins (concrete description of what the variable is)
  - $x = 0, 1, 2, 3$  (what values it could be)
  - This is a discrete, random variable, because  $x$  has                      different values (random) and because those different values follow the discrete rules, i.e. “countable” in the sense that it is a number of times and no intervals between integers

**? Review: What is a synonym for 'fair' when discussing probability?**

- ☒ A) Equally similar  
☐ B) Similar  
☐ C) Equivalent  
☐ D) Equally likely

Explanation:

## 2 Probability Distribution Function (PDF) for a Discrete Random Variable

### 2.1 Introduction

- A **probability distribution function (PDF)** is a very scary way to say the list of possible  $x$  values and their respective                     
  - The                      they follow are simple:
    1. All probabilities must sum up to 1, or  $\sum P(x) = 1$
    2. Each probability is between 0 and 1, inclusive, or  $0 \leq P(x) \leq 1$

 Discuss: Review: try writing the mathematical notation for probability of event A equaling a 30% chance of occurring

- In a probability distribution function, we can think of each                      value of  $X$ , i.e., the  $x$ s, having some specified probability of occurring

### 2.2 PDF Tables

- Example:
  - A survey is being done on how many vehicles families own. Assume the following probability distribution function:
  - $X$  = the number of vehicles owned by a family
  - $x = 0, 1, 2, 3, 4$

| $x$ | $P(X = x)$ |
|-----|------------|
| 0   | 0.10       |
| 1   | 0.20       |

| $x$ | $P(X = x)$ |
|-----|------------|
| 2   | 0.30       |
| 3   | 0.25       |
| 4   | 0.15       |

or...

| $x$ | $P(x)$              |
|-----|---------------------|
| 0   | $P(x = 0) = 10/100$ |
| 1   | $P(x = 1) = 20/100$ |
| 2   | $P(x = 2) = 30/100$ |
| 3   | $P(x = 3) = 25/100$ |
| 4   | $P(x = 4) = 15/100$ |

- In this example, If I were to randomly select a single family from this distribution, I would have a 30% chance of picking a family with 2 cars

### ! Important

These tables should look somewhat familiar - they are similar to the relative frequency tables of variables we looked at in module 2!

 Discuss: On 30% of days I bring one umbrella with me, and on 70% of days I bring no umbrella with me. Write out a PDF table as above to demonstrate this probability distribution function

- There are 4 types of distributions for discrete random variables:
  - Distributions
  - Distributions
  - Distributions
  - Distributions
  - Each of these has their place, but the last 3 come into play more on more  statistical techniques - we'll focus just on the binomial distribution for this lecture

### ! Important

Just because I leave these distribution out in this lecture, doesn't mean that they aren't important! It's mostly just that the others won't be as useful immediately to beginners in statistics.

## 3 Mean or Expected Value and Standard Deviation

### 3.1 Introduction

- With PDFs, we sometimes may wish to find the **expected value/expected long-term average**. Thus, doing running this experiment over and over again, we'd expect to                      on this expected mean
  - The same thing can be done for                      deviation
- This is based in the **law of large numbers** a topic alluded to several times in the previous modules
  - Somewhat review from the                      module: this law states that relative observed frequency approaches the theoretical probability as the number of experiments or trials increases

 Discuss: How did we apply the idea of the law of large numbers previously? What did it mean for samples and populations?

- For a discrete probability function:

$$\mu_{LT} = \sum (x \cdot P(x))$$

$$\sigma_{LT}^2 = \sum [(x - \mu)^2 \cdot P(x)]$$

$$\sigma_{LT} = \sqrt{\sum [(x - \mu)^2 \cdot P(x)]}$$

 Discuss: Why would we use the mu and sigma notation here, instead of x bar and lowercase s? Do they represent statistics or parameters?

### 3.2 Mean Calculation Example

Car example from earlier:

| x | P(x) | x * P(x) |
|---|------|----------|
| 0 | 0.10 | 0.00     |
| 1 | 0.20 | 0.20     |
| 2 | 0.30 | 0.60     |
| 3 | 0.25 | 0.75     |
| 4 | 0.15 | 0.60     |

$$\mu_{LT} = \sum (x * P(x)) = 2.15$$

### 3.3 Standard Deviation Calculation Example

Car example from earlier:

| x | P(x) | x * P(x) | (x - mu)^2 * P(x) |
|---|------|----------|-------------------|
| 0 | 0.10 | 0.00     | 0.462250          |
| 1 | 0.20 | 0.20     | 0.264500          |
| 2 | 0.30 | 0.60     | 0.006750          |
| 3 | 0.25 | 0.75     | 0.180625          |
| 4 | 0.15 | 0.60     | 0.513375          |

$$\sigma_{LT} = \sqrt{\sum (x - \mu)^2 * P(x)} = 1.1948$$

### 3.4 Section Conclusion

- Probability distribution functions serve a purpose - that is - they can describe a particular \_\_\_\_\_ in the probability of outcomes in the data.
  - However, in actual practice, we don't \_\_\_\_\_ the probability distribution function of most variables (i.e., we lack an exact knowledge of the \_\_\_\_\_).
  - Instead, we often aim to apply “ideal” versions of a probability distribution function, in order to examine our data

#### ! Important

The example I showed you above is purposefully reductive; it's to demonstrate the characteristics of the PDF, not something we'd actually do in an analysis

- Once we understand what pattern the variable fits, we can use this information for other \_\_\_\_\_, as there are the 4 specialty distributions mentioned earlier: geometric, hypergeometric, poisson, and the **Binomial Distribution**

## 4 Binomial Distribution

### 4.1 Introduction

- The **binomial distribution** has a certain, \_\_\_\_\_ number of subsequent trials represented as  $n$ 
  - Each trial is \_\_\_\_\_ and does not affect subsequent trials
  - Not every discrete numeric variable is well-suited to be described by the binomial distribution

#### ! Important

Don't forget that the binomial distribution is an 'ideal' - it is useful but not necessarily representative of 'real' data.

- There are two possible \_\_\_\_\_ in the binomial distribution, “success” and “failure”
  - $P(\text{success}) = p$
  - $P(\text{failure}) = q$
  - $p + q = 1$
- The experiment described above fits the \_\_\_\_\_ probability

distribution. Where the random discrete variable  $X$  represents the number of successes obtain in  $n$  trials

- For the binomial probability distribution:

$$\mu_{LT} = n * p$$

$$\sigma_{LT}^2 = n * p * q$$

$$\sigma_{LT} = \sqrt{n * p * q}$$

## 4.2 Notation

$$X \sim B(n, p)$$

- This formula is read as “X is a random variable with a binomial distribution”, where  $n$  and  $p$  represent the same things as before

 Discuss: Why does  $q$  or probability of failures not need to be represented in this notation?

- Example: A child is placed in a new behavioral intervention classroom to better help the child cope with the school environment. On any given day, they have a 40% chance of having a “good” behavior day and a 60% chance of having a “bad” behavior day, measured over 3 subsequent days
  - $X \sim B(0.40, 3)$
  - $X$  = Number of good behavior days (or successes!)
  - $p = P(\text{success}) = 0.40$
  - $q = P(\text{failure}) = 0.60$
  - $n = 3$

## 4.3 Mean Calculation Example

$$\mu_{LT} = 3 * 0.40 = 1.20$$

## 4.4 Standard Deviation Calculation Example

$$\sigma_{LT}^2 = 3 * 0.40 * 0.60 = 0.72$$

$$\sigma_{LT} = \sqrt{3 * 0.40 * 0.60} = 0.85$$

## 5 Conclusion

### 5.1 Recap

- In this shorter lecture, we re-introduced the concept of discrete random variables, and how they can be represented with probability distribution functions
- We played around with calculating expected mean and standard deviations for outcomes from the probability distribution function table
- We introduced the binomial distribution as a special, specific pattern for a probability distribution represented via “successes” and “failures”

## Key Terms

### B

**binomial distribution** An ideal probability distribution for discrete variables, involving a fixed number of independent, subsequent trials 8

### D

**discrete variable** A numeric variable that is represented in counts 2

### E

**expected value/expected long-term average** In a probability distribution function, a way of describing the 'average' value, if a value from the distribution was taken an infinite number of times 6

### L

**law of large numbers** Applied to probability distribution functions, it describes how, as a sample grows larger, the mean average converge on a single point 6

### P

**probability distribution function (PDF)** A representation of the different possible values of a random discrete variable, and the relative probabilities of each value 4

## R

**random variables** A variable that can take different values 3

*The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in your book and your own judgment to make sure you have all relevant terms*